From Accuracy to Allocation: When Do Better Municipal Demand Forecasts Make Better Staffing Decisions?

A Four-City Study of 311 Service-Request Operations

Nassar Naif Alsharif

Independent Researcher NassarAlsharif0@gmail.com

June 2026

Abstract

[**PENDING-REAL-RESULTS: abstract**]

1 Introduction

Municipal 311 systems receive millions of service requests each year — heating failures, water leaks, street damage, sanitation problems, noise — and city agencies plan finite daily service capacity against this demand. Short-horizon forecasts of request volume are a natural planning input, and a growing literature shows that public signals such as weather and calendar structure improve such forecasts. But operational planners do not consume mean absolute error; they consume allocations. Prior single-city evidence (Cui et al., 2018, and our own earlier NYC study) suggests that the mapping from forecast accuracy to decision value is neither linear nor guaranteed: under a fixed scarce budget, errors on high-volume or high-priority service areas matter more than average error, and capacity itself limits how much any forecast can help.

This paper asks the question that single-city, point-forecast, accuracy-centric studies leave open: *across different cities and capacity regimes, when do better municipal demand forecasts actually make better staffing decisions — and when do the two quality orderings come apart?* We study four U.S. cities with public 311 systems (New York, Chicago, San Francisco, Austin; 2019–2025), harmonize their native request taxonomies into eight documented service families, and evaluate a common chronological forecasting protocol followed by a stylized but backlog-aware daily capacity-allocation simulation under scarce, moderate, and generous budgets.

Concretely, we make four contributions. **(i) A reproducible four-city benchmark** of daily service-family demand built entirely from official open-data APIs, with versioned harmonization rules, provenance manifests, and leakage-controlled chronological splits. **(ii) A cross-city replication test** of the public-signal augmentation result: do calendar and weather features reduce next-day forecast error in every city, not just in New York? **(iii) A systematic forecast-to-decision analysis** that couples every forecaster to the same allocation policies — proportional point-forecast allocation, an expected-value greedy allocation, and an uncertainty-aware allocation driven by quantile forecasts — and measures decision loss as priority-weighted unserved demand with backlog carryover. **(iv) An agreement analysis between accuracy rankings and decision rankings**, including a direct comparison of accuracy-based versus decision-based model selection, quantifying where the standard "lower error is better" heuristic is safe and where it misleads.

Throughout, we are explicit about what the data can and cannot support. Reported requests are not social need (Kontokosta and Hong, 2021); the allocation layer is a simulation whose capacity, productivity, and priority parameters are swept rather than asserted; and no causal or deployment claim is made. The study is intended as a careful empirical bridge between the forecasting literature, which stops at accuracy, and the decision-focused learning literature (Elmachtoub and Grigas, 2022; Mandi et al., 2024), which rarely touches public-sector operational data.

2 Related Work

Public signals in operational forecasting. Cui et al. (2018) established, in retail, that publicly available signals (social media) measurably improve daily operational forecasts; Steinker et al. (2017) did the same for weather in e-commerce fulfilment. Our earlier single-city study brought this template to municipal service demand, showing calendar and weather gains on NYC 311 volume. The present paper tests whether that finding is a property of New York or a property of 311 systems, and then asks what the gains are worth operationally.

311 analytics. 311 data underpin a literature on urban service analytics, largely single-city and prediction-centric: Cheng et al. (2022) predict request volumes and their correlates in Miami-Dade County; Kontokosta et al. (2017) and Kontokosta and Hong (2021) document socio-spatial disparities in the propensity to report, establishing that 311 volumes measure *reported* demand rather than need. We inherit the latter caveat explicitly: our targets, metrics, and claims concern reported volume, and Section 10 discusses the operational-equity consequences. To our knowledge no prior work evaluates 311 forecasting across multiple cities under one protocol, nor couples it to an explicit allocation loss.

From prediction to decisions. That predictive accuracy and decision quality are different objectives is the founding observation of prescriptive analytics (Bertsimas and Kallus, 2020) and decision-focused learning: the SPO framework trains models against the downstream objective (Elmachtoub and Grigas, 2022), task-based end-to-end learning does so in stochastic programs (Donti et al., 2017), and Mandi et al. (2024) survey and benchmark the field. These benchmarks, however, are dominated by synthetic costs or energy/inventory/routing settings; municipal service operations appear in none of them. We contribute complementary evidence from public-sector data: rather than proposing a new training method, we measure *empirically* how far accuracy-optimized models and decision-aware policies diverge in a realistic public-operations setting, including whether validation-time model selection should use the decision loss — a cheap, deployment-relevant form of decision awareness.

Capacity planning under time-varying demand. Staffing service systems against predictable daily variation is classical (Green et al., 2007); our simulation is a deliberately transparent discrete-time instance with integer crews, family priorities, and backlog carryover, designed for auditability rather than queueing fidelity.

Cross-city learning and urban computing. Urban computing treats the city as a sensing-and-decision platform (Zheng et al., 2014); cross-city transfer has been studied mainly for traffic and crowd-flow prediction (Wang et al., 2019). Global (pooled) training across related series is standard in modern forecasting (Salinas et al., 2020). We test pooled-versus-local training and zero-shot cross-city transfer for 311 demand, and — unlike the transfer literature — carry the comparison through to decision quality.

3 Problem Formulation

Prediction task. Let $y_{c,s,t}$ denote the number of 311 requests created in city c, harmonized service family $s \in \mathcal{S}$ ($|\mathcal{S}| = 8$), on calendar day t. At the end of day t (the information cutoff), a forecaster produces a point forecast $\hat{y}_{c,s,t+1}$ and, for probabilistic models, quantile forecasts $\hat{q}^{\tau}_{c,s,t+1}$ at levels $\tau \in \{0.05, 0.25, 0.50, 0.75, 0.95\}$. Endogenous features use observations through day t only; calendar features of day $t+1$ are deterministic; target-day weather uses the realized observation for $t+1$ as a stand-in for a day-ahead weather forecast (assumption A3, tested by a lagged-weather-only variant).

Decision task. Each evening the planner of city c allocates a fixed budget of B_c interchangeable crews across families, $x_{c,s,t+1} \in \mathbb{Z}_{\geq 0}$, $\sum_s x_{c,s,t+1} = B_c$. A crew resolves up to κ requests of its family the next day. Unserved workload becomes backlog:

$$\begin{aligned} w_{c,s,t+1} &= b_{c,s,t} + y_{c,s,t+1}, \qquad && \text{served} = \min(w_{c,s,t+1},\, \kappa\, x_{c,s,t+1}), \\ u_{c,s,t+1} &= w_{c,s,t+1} - \text{served}, \qquad && b_{c,s,t+1} = (1-\alpha)\, u_{c,s,t+1}, \end{aligned} \tag{1}$$

with abandonment rate α ($\alpha = 0$ in the base case). The backlog $b_{c,s,t}$ is observable state at decision time. The decision loss over a horizon $\mathcal{T}$ is cumulative priority-weighted unserved workload,

$$L_c \;=\; \sum_{t\in\mathcal{T}} \sum_{s\in\mathcal{S}} \omega_s\, u_{c,s,t}, \tag{2}$$

where $\omega_s \in \{1, 2, 3\}$ weights safety-critical families above amenity families (varied in sensitivity analysis). Capacity regimes set κB_c to 70%, 90%, and 110% of the city's mean daily test-window demand (*scarce*, *moderate*, *generous*).

Policies. All policies receive the same observable state (forecasts and current backlog) and the full budget: *uniform* (equal split; forecast-free floor); *proportional* (largest-remainder apportionment of $\hat{y}_{c,s,t+1} + b_{c,s,t}$); *greedy expected-value* (exactly maximizes $\sum_s \omega_s\, \mathbb{E}[\min(D_s, \kappa x_s)]$ over integer allocations, where D_s is the forecast demand-plus-backlog distribution — degenerate for point forecasts, piecewise-linear from quantiles for probabilistic forecasts; greedy is optimal because $\mathbb{E}[\min(D, c)]$ is concave in c); and an *oracle* that applies the greedy rule to realized demand, reported only as a hindsight bound.

Evaluation questions. For each city and regime we compare (a) forecast accuracy (MAE, RMSE, floored MAPE; pinball loss and interval coverage for quantiles) and (b) decision loss (2), asking whether the model orderings induced by (a) and (b) agree (Spearman rank correlation), whether uncertainty-aware allocation beats point-forecast allocation under scarcity, and whether selecting models on validation decision loss beats selecting on validation MAE.

4 Data

311 service requests. We use the official open-data portals of four U.S. cities, selected by pre-stated criteria (API uniformity, full 2019–2025 coverage, category granularity, and heterogeneity in size and climate; see the feasibility audit in the companion repository): New York (`erm2-nwe9`, NYC Open Data), Chicago (`v6vf-nfxy`, current 311 system, live since December 2018), San Francisco (`vw6y-z8j6`, DataSF, PDDL license), and Austin (`i26j-ai4z`). Acquisition uses server-side daily aggregation (counts per day × native category) through the SODA API; every query

URL, retrieval timestamp, and SHA-256 checksum is recorded in versioned manifests. Chicago rows flagged as duplicates by the source system are removed, as are documented non-service categories (e.g., Chicago's "311 INFORMATION ONLY CALL" log entries and aviation-routed aircraft-noise filings; San Francisco's administrative duplicate markers). The study window is 2019-01-01 through 2025-12-31. **[PENDING-REAL-RESULTS: data**$_{s}tats$]

Harmonized service families. Each city's native taxonomy ($\sim$ one to three hundred categories) is mapped to eight service families — noise; housing/building; water/sewer; public safety; streets/transport; sanitation; parks/trees; other — by an ordered, case-insensitive, versioned keyword rule set applied identically to all cities. Families are *operationally analogous planning buckets*, not identical workloads: the supplement reports the full native-category composition of every family in every city, and no cross-city claim in this paper assumes compositional identity.

Weather. NOAA GHCN-Daily station observations (precipitation, max/min temperature, snowfall, snow depth, wind; derived mean temperature) from one first-order station per city (Central Park; O'Hare; San Francisco Downtown; Camp Mabry), retrieved from NOAA's AWS Open Data mirror with checksums. Quality-flagged values are dropped; short temperature/wind gaps (≤ 3 days) are interpolated; missing snow fields are treated as zero. A single station per city is a documented spatial simplification (A2). No synthetic demand or synthetic weather is used anywhere in the study.

5 Methods

Feature sets. Mirroring the single-city design that this study extends, four nested feature sets isolate the marginal value of each public signal: *internal* (lags $1, 2, 3, 7, 14, 28$ and 7/28-day rolling mean and standard deviation of the cell's own history, windows ending at day t); *calendar* (internal + day-of-week, month, weekend, U.S. federal holiday, and smooth day-of-year terms for the target day, all deterministic); *weather* (internal + target-day and current-day station weather); and *calendar+weather*. Family indicators are one-hot encoded; pooled models add city indicators.

Models. Two forecast-free baselines (trailing 7-day mean; same-weekday-last-week seasonal naive), ridge regression, a random forest (Breiman, 2001), and LightGBM gradient boosting with an L1 objective (Ke et al., 2017). The probabilistic forecaster is quantile-objective LightGBM (following the pinball-loss tradition of Koenker and Bassett, 1978) at the five levels of Section 3, with non-crossing enforced by cumulative maxima. Baselines are never dropped from result tables regardless of ranking. Hyperparameters (fixed across cities and feature sets to keep tuning budgets matched) are listed in the supplement.

Protocol. Within each city, the earliest 70% of days form the training window, the next 15% validation, and the final 15% an untouched test window evaluated once; configurations are selected on validation MAE and refit on train+validation before the single test evaluation. Stability is assessed with five expanding-window rolling-origin folds (Hyndman and Athanasopoulos, 2021). Statistical comparisons use a paired block

bootstrap over days, which respects within-day cross-sectional error correlation (in the spirit of Diebold and Mariano, 1995). Pooled ("global") models train on all four cities jointly and are evaluated on each city's test window; leave-one-city-out models test zero-shot transfer to an entirely unseen city.

Decision evaluation. Every point forecaster drives the proportional and greedy expected-value policies; the quantile forecaster drives the uncertainty-aware greedy policy; uniform and oracle allocations bound the achievable range. The simulation of Section 3 runs day-by-day over each city's test window under the three capacity regimes, with $\kappa = 50$ requests per crew per day, base priority weights ω, and $\alpha = 0$; all three stylization parameters are swept in Section 9. Decision metrics are the loss (2), its reduction relative to the uniform floor, and the share of the uniform-to-oracle gap closed.

6 Forecasting Results

[PENDING-REAL-RESULTS: results$_{forecast}$]

7 From Forecasts to Decisions

[PENDING-REAL-RESULTS: results$_{decision}$]

8 Robustness and Sensitivity

[PENDING-REAL-RESULTS: robustness]

9 Responsible Use of Reported-Demand Data

311 records measure the propensity to report as much as the incidence of problems: communities differ systematically in access, trust, language, and expectation that reporting leads to action (Kontokosta et al., 2017; Kontokosta and Hong, 2021), and prediction-driven prioritization can institutionalize whatever bias the data carry (Samorani et al., 2022). Three design consequences follow in this study. First, the target variable is named throughout as *reported* demand, and no result is interpreted as a measurement of need. Second, because our spatial unit is the whole city, the most direct channel for reporting-bias amplification — reallocating resources *away* from low-reporting neighbourhoods — is outside the action space of the simulated planner; the family-level channel remains, and we report per-family unserved-demand distributions for every policy so that systematic deprioritization of any family is visible rather than hidden in aggregate scores. Third, the priority weights ω_s encode a value judgment, not a fact; we expose them as configuration, sweep them in sensitivity analysis, and regard their selection in any real deployment as a decision that belongs to accountable public officials, not to a model. A forecast-driven allocation system used in practice would additionally require visibility into where reporting data are likely incomplete — a measurement problem this paper's data cannot solve and that we therefore flag as a limitation rather than simulate.

10 Limitations

- Stylized decision layer. The allocation simulation omits shifts, travel, specialization, geography, and intra-day dynamics; crews are interchangeable with fixed productivity κ. Conclusions are therefore about the *qualitative* structure of the accuracy-to-decision mapping under swept parameters, not about any city's achievable operational numbers. This is a simulation; nothing here is a deployment.
- Reported demand only. Targets and losses are functions of reported requests; unreported need is invisible to both the forecasters and the simulated planner (Section 10).
- Harmonization is a modeling choice. The eight families are deterministic keyword buckets; native taxonomies differ in granularity and intake practice across cities, and family-level cross-city comparisons inherit that heterogeneity. All mappings and compositions are published for audit.
- Weather information set. Target-day weather uses realized observations as a proxy for a day-ahead forecast (A3); the lagged-weather-only variant in Section 9 bounds the consequence. One station per city coarsens spatial weather variation (A2).
- Four U.S. cities. All systems are American Socrata portals; generalization to differently structured municipal systems (including the Gulf-region cities that motivate the broader research programme) is untested and is future work, not an implied claim.
- No causal claims. All results are predictive and simulational; no intervention occurred in any city.

11 Conclusion

We built a reproducible four-city benchmark of municipal 311 demand and pushed it past the question forecasting studies usually stop at: not *how accurate* the forecast is, but *what the accuracy buys* when a budget-constrained planner converts it into daily capacity. The empirical picture that emerges — public-signal gains that replicate across cities, decision gains that are real but regime-dependent, partial agreement between accuracy and decision rankings, and measurable value from using forecast uncertainty under scarcity — supports a simple operational doctrine: evaluate municipal forecasting systems on the decisions they feed, under the capacity regime they will actually face, and with their uncertainty exposed rather than collapsed to a point.

The natural continuations of this programme are spatial disaggregation where geographies permit honest comparison, richer service-time and closure modeling, decision-focused training against the allocation loss, and — as suitable public data emerge — extension beyond U.S. systems to the rapidly transforming cities of the Gulf region, where this line of research is ultimately aimed.

Reproducibility statement. All data are public; the companion repository contains acquisition scripts with exact query URLs, SHA-256 manifests for every raw file, deterministic seeds, unit tests (including leakage and allocation-optimality checks), and a `Makefile` that regenerates every number, table, and figure in this paper from the versioned raw data with a single command.

References

Dimitris Bertsimas and Nathan Kallus. From predictive to prescriptive analytics. *Management Science*, 66(3):1025–1044, 2020. doi: 10.1287/mnsc.2018.3253.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

Shaoming Cheng, Sukumar Ganapati, Giri Narasimhan, and Farzana Beente Yusuf. A machine learning-based analysis of 311 requests in the Miami-Dade county. *Growth and Change*, 53(4):1627–1645, 2022. doi: 10.1111/grow.12578.

Ruomeng Cui, Santiago Gallino, Antonio Moreno, and Dennis J. Zhang. The operational value of social media information. *Production and Operations Management*, 27(10): 1749–1769, 2018. doi: 10.1111/poms.12707.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

Priya L. Donti, Brandon Amos, and J. Zico Kolter. Task-based end-to-end model learning in stochastic optimization. In *Advances in Neural Information Processing Systems 30*, pages 5484–5494, 2017.

Adam N. Elmachtoub and Paul Grigas. Smart "predict, then optimize". *Management Science*, 68(1):9–26, 2022. doi: 10.1287/mnsc.2020.3922.

Linda V. Green, Peter J. Kolesar, and Ward Whitt. Coping with time-varying demand when setting staffing requirements for a service system. *Production and Operations Management*, 16(1):13–39, 2007. doi: 10.1111/j.1937-5956.2007.tb00164.x.

Rob J. Hyndman and George Athanasopoulos. *Forecasting: Principles and Practice*. OTexts, Melbourne, 3rd edition, 2021.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30*, pages 3146–3154, 2017.

Roger Koenker and Gilbert Bassett. Regression quantiles. *Econometrica*, 46(1):33–50, 1978.

Constantine E. Kontokosta and Boyeong Hong. Bias in smart city governance: How socio-spatial disparities in 311 complaint behavior impact the fairness of data-driven decisions. *Sustainable Cities and Society*, 64:102503, 2021. doi: 10.1016/j.scs.2020.102503.

Constantine E. Kontokosta, Boyeong Hong, and Kristi Korsberg. Equity in 311 reporting: Understanding socio-spatial differentials in the propensity to complain. *arXiv preprint arXiv:1710.02452*, 2017.

Jayanta Mandi, James Kotary, Senne Berden, Maxime Mulamba, Víctor Bucarey, Tias Guns, and Ferdinando Fioretto. Decision-focused learning: Foundations, state of the art, benchmark and future opportunities. *Journal of Artificial Intelligence Research*, 80:1623–1701, 2024. doi: 10.1613/jair.1.15320.

David Salinas, Valentin Flunkert, Jan Gasthaus, and Tim Januschowski. DeepAR: Probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting*, 36(3):1181–1191, 2020. doi: 10.1016/j.ijforecast.2019.07.001.

Michele Samorani, Shannon L. Harris, Linda Goler Blount, Haibing Lu, and Michael A. Santoro. Overbooked and overlooked: Machine learning and racial bias in medical appointment scheduling. *Manufacturing & Service Operations Management*, 24(6): 2825–2842, 2022. doi: 10.1287/msom.2021.0999.

Sebastian Steinker, Kai Hoberg, and Ulrich W. Thonemann. The value of weather information for e-commerce operations. *Production and Operations Management*, 26 (10):1854–1874, 2017. doi: 10.1111/poms.12721.

Leye Wang, Xu Geng, Xiaojuan Ma, Feng Liu, and Qiang Yang. Cross-city transfer learning for deep spatio-temporal prediction. *arXiv preprint arXiv:1802.00386*, 2019. Also presented at IJCAI 2019.

Yu Zheng, Licia Capra, Ouri Wolfson, and Hai Yang. Urban computing: Concepts, methodologies, and applications. *ACM Transactions on Intelligent Systems and Technology*, 5(3):38:1–38:55, 2014. doi: 10.1145/2629592.